Modern Physics (PHY 211)
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Dr. Samalka Anandagoda

Chapter 1: The Birth of Modern Physics

1.1 Classical Physics of the 1890s

- There are two major branches of Physics:
  - **Classical Physics**
  - **Modern Physics**

- Classical Physics includes:
  - Newton's Laws of mechanics
  - Gravitation
  - Electricity and Magnetism which have been unified by Maxwell's theoretical work
  - Electromagnetic Waves – predicted by Maxwell's equations and discovered and investigated by Hertz
  - The laws of thermodynamics and kinetic theory

1.1 Classical Physics of the 1890s

- Classical theories were successful in explaining the macroscopic world. That is the world we see.

- Scientists used this knowledge to build engines, ships, equipment and use them for human advantage.

Mechanics

- Many researchers contributed to develop the laws of Mechanics. Issac Newton (1642-1727) is known as the greatest scientist of his time.
- His discoveries were in the fields of mathematics, astronomy, and physics.

It may not be so apparent to us today, but we should not forget the tremendous unification that Newton made when he pointed out that the motions of the planets about our sun can be understood by the same laws that explain motion on Earth, like apples falling from trees or a soccer ball being shot toward a goal.

(Thornton, S.T., Rex, A and Hood, C., Modern Physics for Scientists and Engineers. Fifth Edition, Cengage).

Newton's three Laws of Motion

- Newton's first law:
  - *An object at rest will stay at rest and an object in motion with a constant velocity will continue in motion unless acted upon by some net external force.*

- Newton's Second Law:

$$\vec{F} = m\vec{a}$$

  - *The acceleration* $\vec{a}$ *of a body is proportional to the net external force F and inversely proportional to the mass m of the body.*

- Newton's Third Law:

$$\vec{F}_{21} = -\vec{F}_{12}$$

  - *The force exerted by body 1 on body 2 is equal in magnitude and opposite in direction to the force that body 2 exerts on body 1.*

Electromagnetism

- Few notable researchers:
  - Charles Coulomb (1736– 1806)
  - Hans Christian Oersted (1777–1851)
  - Thomas Young (1773– 1829)
  - André Ampère (1775– 1836)
  - Michael Faraday (1791– 1867)
  - Joseph Henry (1797– 1878)
  - James Clerk Max well (1831– 1879)
  - Heinrich Hertz (1857– 1894).

Coulomb Law :

$$F = k\frac{q_1 q_2}{r^2}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

Maxwell’s equations

- Gauss’s law for electricity:

$$\oint \vec{E}.d\vec{A} = \frac{q}{\epsilon_0}$$

- Gauss’s law for magnetism:

$$\oint \vec{B}.d\vec{A} = 0$$

- Faraday’s law:

$$\oint \vec{E}.d\vec{s} = -\frac{d\phi_B}{dt}$$

- Generalized Ampere’s law:

$$\oint \vec{B}.d\vec{s} = \mu_0\epsilon_0\frac{d\phi_E}{dt} + \mu_0 I$$

- Lorentz force law:

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

1.2 The Kinetic Theory of Gases

Kinetic theory of gasses tells us how Pressure, Temperature and Volume are related to each other. Two key laws are Boyle's Law and Charles Law (also known as Gay-Lussac's Law).

- Boyle's Law:

  Pressure times Volume of an ideal gas is a constant

$$PV = k$$

- Charles's Law:

  Volume divided by Temperature is a constant

$$\frac{V}{T} = k$$

- Combining these two laws we get the Ideal Gas equation:

$$PV = n\,R\,T$$

Why need new Laws?

- Classical physical laws are able to explain phenomena at macroscopic level.
- Can these laws also explain phenomena at a microscopic level? Can Newtonian mechanics explain the movements of extremely small objects?
- Can classical laws explain the motion of objects moving at speeds closer to the speed of light?

At atomic and subatomic scales, classical laws like Newton's, Maxwell's, and classical thermodynamics fail to give accurate predictions. Also, Classical physics fails for objects moving at speeds close to the speed of light.

Practice Questions

1. Two point charges, q_1=+3.0 C and q_2 = -2.0 C are placed 0.5 m apart in vacuum. The Coulombic force between the charges are given by $F = \frac{kq_1q_2}{r^2}$, where k =8.98x10^9 Nm^{2}C^{-2}

   a). What is the magnitude of the force between them.

   b). Is the force attractive or repulsive.

   c). What is the Force between the charges if the distance between them is doubled?

2. A gas has a volume of 4.0 L at a pressure of 200 kPa. If the **temperature remains constant**, what will the volume be when the pressure is increased to 500 kPa? (use Boyle's Law)